Zhang 2019 HGSOC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates both bilateral omental sites of patient VOA11543: same-sample neighbour fraction 0.997→0.641 (perfect-mix floor 0.559), iLISI median 1.00→1.78 of 2. One connected component after removing a 20-cell island. Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

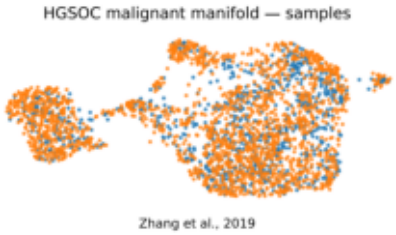
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom do NOT compete out of a hub (smoothed r=−0.10), both oppose Crest (pSom↔Crest r=−0.64, Gast↔Crest r=−0.44). Along pseudotime Gast falls (ρ −0.50), Crest rises to a terminal tip (ρ +0.34, decile pt=0.80). Joint pSom&Crest enrichment 0.35 (<1 = inverse gradient).

Gate 3 — cross-repression → NA

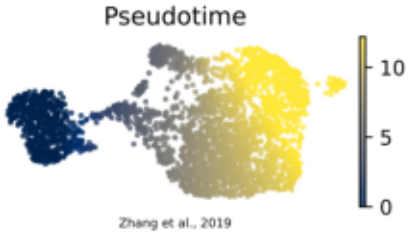
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run (correlation / modules notebooks are not produced).

STEP FIGURES

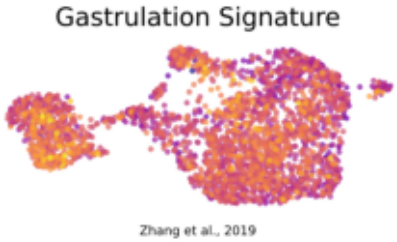
Manifold — samples



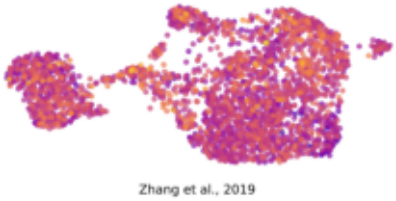
Pseudotime



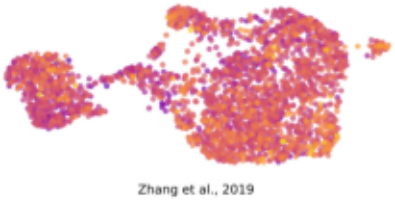
Gastrulation



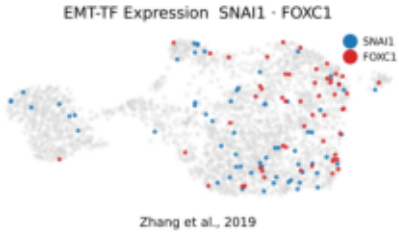
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
HGSOC bilateral ovarian tumour (Zhang et al., 2019), 10x — 2,782 malignant cells over 2 omental sites (1 patient, VOA11543 left+right ovary); predominantly low-cycling (~9% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1 (root-side) → FOXC1 (terminus).

Parameter choices

Sample threshold	≥100-cell floor; 2 bilateral sites (no cliff) → 2/2, 2,802 cells	**Overdispersed genes**	1,247 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (bilateral-site batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 20-cell island removed
Geometry	non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)	**Root**	tip 0 = Gast/pSom-high, Crest-low origin
Module means (raw)	Gast −0.011 · pSom +0.042 · Crest −0.034 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (root-side) → FOXC1 (terminus); FOXC/PRRX in panel